than we or our equipment can detect. But let’s now reexamine this issue in the braneworld scenario. How do we detect things? Well, when we use our eyes, we use the electromagnetic force; when we use powerful instruments like electron microscopes, we also use the electromagnetic force; when we use atom smashers, one of the forces we use to probe the ultrasmall is, once again, the electromagnetic force. But if the electromagnetic force is confined to our three-brane, our three space dimensions, it is *unable* to probe the extra dimensions, regardless of their size. Photons cannot escape our dimensions, enter the extra dimensions, and then travel back to our eyes or equipment allowing us to detect the extra dimensions, *even if they were as large as the familiar space dimensions.*

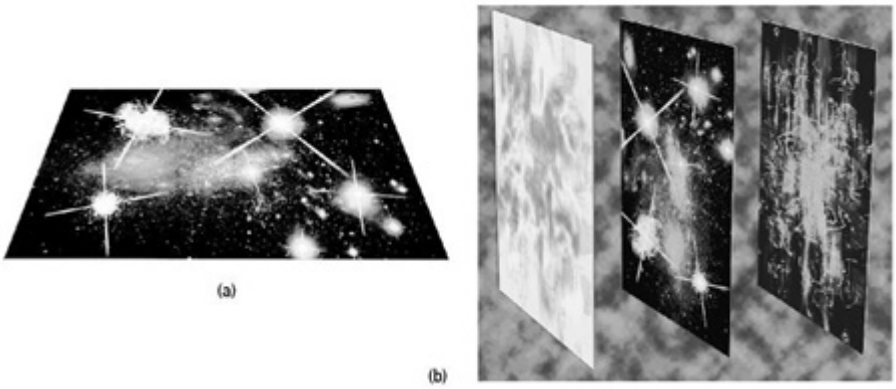


Figure 13.3 **(**a**)** In the braneworld scenario, photons are open strings with endpoints trapped within the brane, so they—light—cannot leave the brane itself. **(**b**)** Our braneworld could be floating in a grand expanse of additional dimensions that remain invisible to us, because the light we see cannot leave our brane. There might also be other braneworlds floating nearby.

So, if we live in a three-brane, there is an alternative explanation for why we’re not aware of the extra dimensions. It is not necessarily that the extra dimensions are extremely small. They could be big. We don’t see them because of the *way* we see. We see by using the electromagnetic force, which is unable to access any dimensions beyond the three we know about. Like an ant walking along a lily pad, completely unaware of the deep waters lying just beneath the visible surface, we could be floating within a grand, expansive, higher-